

Figures for Academic Journal Publications

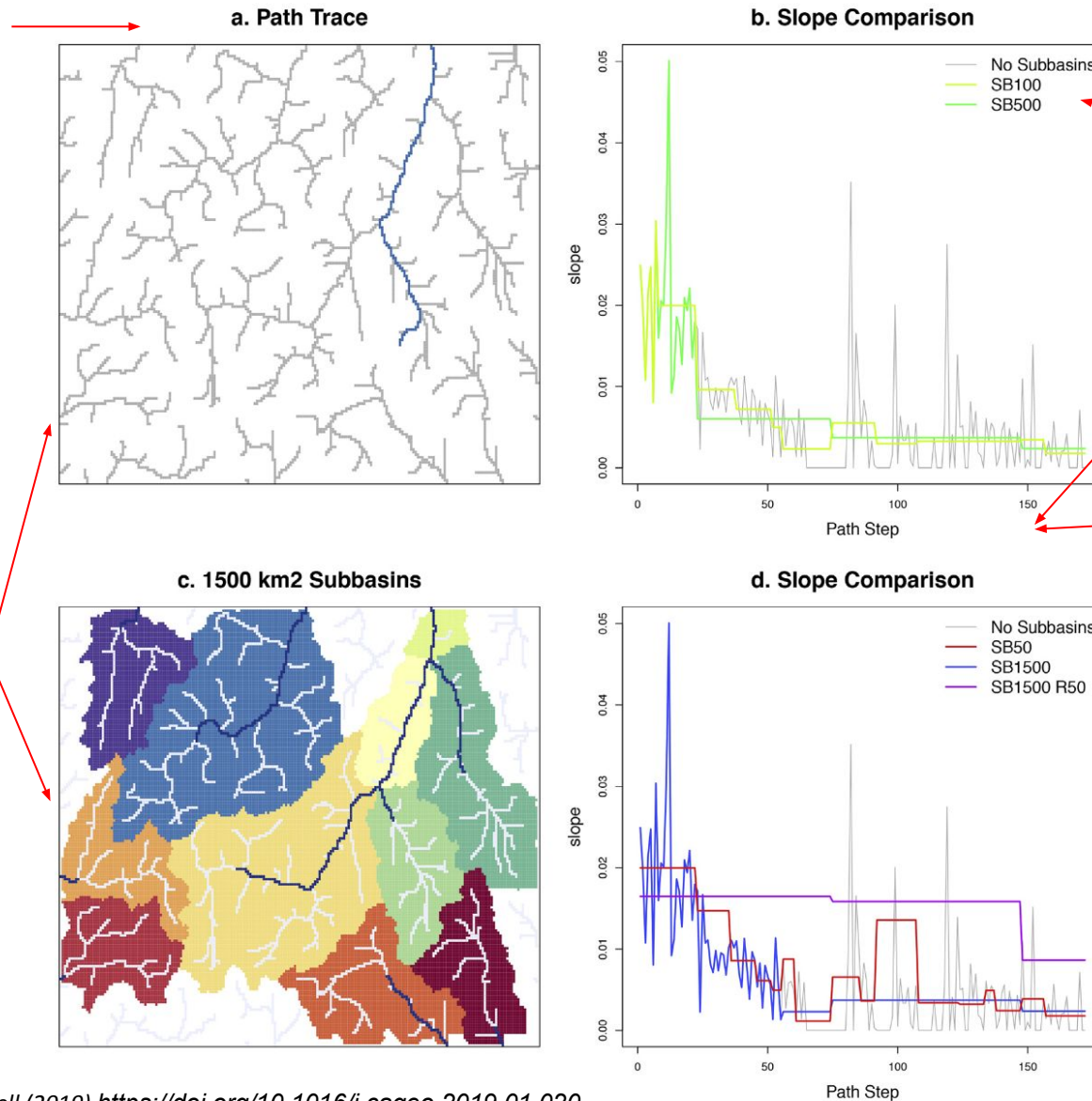
Condon Research Group meeting 2/15/24

Anatomy of a publication ready figure

Every subplot has a letter title (short subtitles are optional)

All subplot titles are in the same font and same font size (this can be a different font size than the axes and legends but should be the same font).

All subplots are vertically and horizontally aligned with even spacing between them



There is no main title for the figure

A legend is provided as needed to define all necessary symbology but the same legend is never repeated more than once

All axes are labeled, and all labels include units if relevant

Axes and annotation fonts are no smaller than size 9 when the figure is sized to fit into a word document (You can test this with a simple text box in ppt for comparison)

A caption is provided which clearly explains all subplots. (NOTE: for some journal articles longer captions are required that also explain the findings of the figure)

Figure x: Slope magnitudes walking down an example path (subplot a) compared between the no-subbasin case and the five subbasin approaches (b and d). Subplot c illustrates the basin delineation using an area threshold of 1500 m. The dark stream pixels in c, show the stream network using a 1500 m threshold and the light are the stream network using a 50 m threshold.

Figure checklist: Use of Color

1. Are you using a sequential color scale if you are dealing with a sequential quantity?
2. Are you using diverging color scale if you have both positive and negative numbers? Have you centered the color scale so that zero is in the center?
3. If you have categorical values are you using color to help highlight relationships (e.g. if you have three groundwater variables and three snow variables you could use purples for one group and greys for another)?
4. Are the colors you are using in this figure consistent with color choices you make in other parts of your paper? (e.g. always using green for groundwater and blue for surface water)
5. Do your colors make topical sense (e.g. using orange for drier conditions and green for wetter)?
6. Are your colors accessible to a broad audience? (e.g. check that they are color blind friendly and that you have maintained a minimal contrast level)
7. Are you using colors of different intensity in an intentional way? If some of your colors are much bolder or more intense than the others they will stand out. If you are doing this on purpose that's fine if not consider making a change.
8. When you look at your figure are the most important things you want to emphasize jumping out at you or are there distracting elements?
9. Could you use less color throughout the figure and instead use color more sparingly to highlight important points?

Figure checklist: Symbolology

1. Are you varying color, shading type, and symbol at the same time to differentiate one thing? (e.g. using a solid blue line for snowpack and a dashed purple line for precipitation in this case both color and line type are being used to differentiate snow from precipitation. If snow was modeled and precipitation was measured this would be okay because you could say the color differentiates the variable (snow vs precipitation) and the line type differentiates whether its measured or modeled)
2. Do you use consistent symbolology across the different figures in your paper?
3. Are your symbols (or line types) big enough to tell them apart when the figure is sized for publication?
4. Can you simplify or reduce the number of symbols you are using?

Figure checklist: Layout and text

**NOTE for all of the following tests you should have your figure scaled to the way it might appear in a publication (i.e. at maximum 7.5" wide)*

1. Does your figure scale nicely to a page or half page without a lot of extra white space? (e.g. when you layout your plots if you have 7 you might want to do three rows of 2 with a blank slot for a legend not a row of 6 and a row of 1)
2. Do you have consistent spacing between all of your subplots?
3. Is every axis labeled and does every label include units?
4. Do you have a legend for everything that needs a legend?
5. Is that legend unnecessarily repeated in subplots? (If subplots have the same information you can just have the legend once)?
6. Does every subplot have a letter title (i.e. a-e)?
7. Is there any font on your figure that is smaller than size 9 when you size it for publication?
8. Do you have unnecessary outlines or lines in your layout? Experiment with getting rid of boxes around legends or other features to see if your figure looks less busy.
9. Are there annotations you could add to your plot that would make it more intuitive? Do you have room to spell these out or do you need to use acronyms?

Figure checklist: Maps

1. Does your map include a scale bar and a north arrow?
2. If its not obvious where the map location is, have you included an inset map showing the location of your map in a wider setting? If appropriate have you included a lat/lon grid?
3. Have you controlled for aspect ratio in any figure manipulations so your map does not appear stretched?
4. If you have more than one map, do you have a consistent basemap or set of labeled features so that readers can easily compare across maps?
5. Is your choice of basemap or reference features distracting? You want to include geographic information to help orient the reader without distracting from your main points.

Accessibility Checklist and Best Practices

- Add alt text to all images of charts, diagrams, and graphs
- Check color contrast ratios
- Run all graphs through a colorblind simulator
- When possible, don't rely on color alone to convey information
- Include data tables with graphics when possible
 - This can be in supplemental information

alt= "**Chart type** of **type of data** where **reason for including chart**"

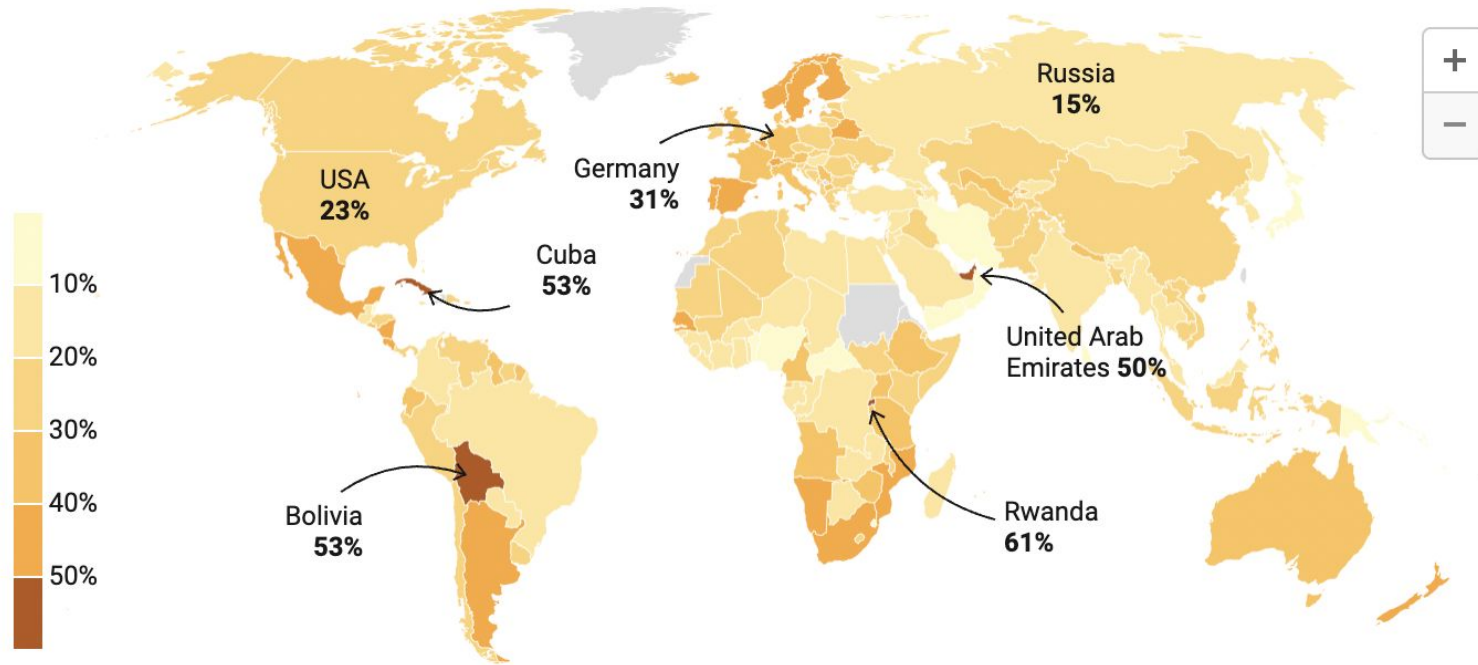
Include a **link to data source** somewhere in the text

[*Amy Cesal: Writing Alt Text for Data Visualization*](#)

Alt text example

Share of seats held by woman in national parliaments

The chart shows the proportion of women in the seats in national parliaments or in the lower house in bicameral systems (in %) in 2020.



Map: Daniela Haake • Source: [Inter-Parliamentary Union \(IPU\)](#)

Alt text:

A choropleth map of the proportion of women in national parliaments in the world in 2020, where Rwanda stands out as the country with the highest percentage of women (61%) in a national parliament, followed by Bolivia and Cuba, with a 53%. Papua New Guinea is the country that scored the worst with 0% of women in their parliament.

Color Contrast Ratios

Type of content	Minimum ratio (AA rating)	Enhanced ratio (AAA rating)
Body text	4.5 : 1	7 : 1
Large-scale text (120-150% larger than body text)	3 : 1	4.5 : 1
Active user interface components and graphical objects such as icons and graphs	3 : 1	Not defined

Good contrast

Contrast Ratio

5.09:1

Bad contrast

Contrast Ratio

1.4:1

Resources for making better figures

Contrast:

- [Information on best ratios](#)
- [Color Contrast Checker](#)

Color Blind Friendly Color Generator:

- [Coloring for Color Blindness](#)

Color Blind Simulator:

- [Coblis](#)

Screen Reader Chrome Extensions:

- [Screen Reader](#)

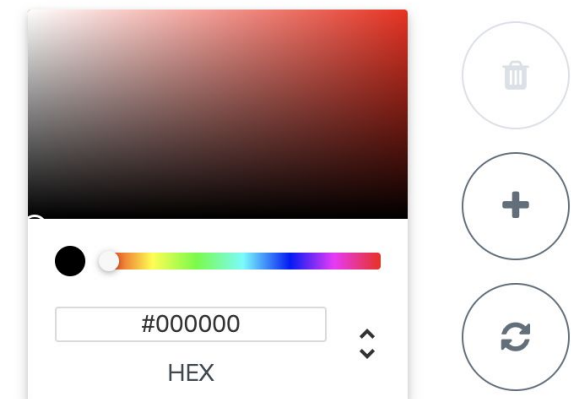
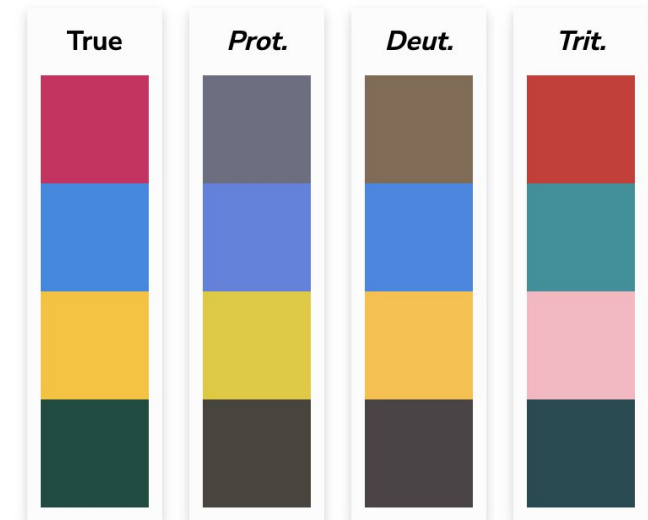
Writing Alt Text:

- [Writing Alt Text for Data Visualization](#)

Deeper Dive into Accessible Science Communication:

- [Presentation by Patrick Garvin](#)

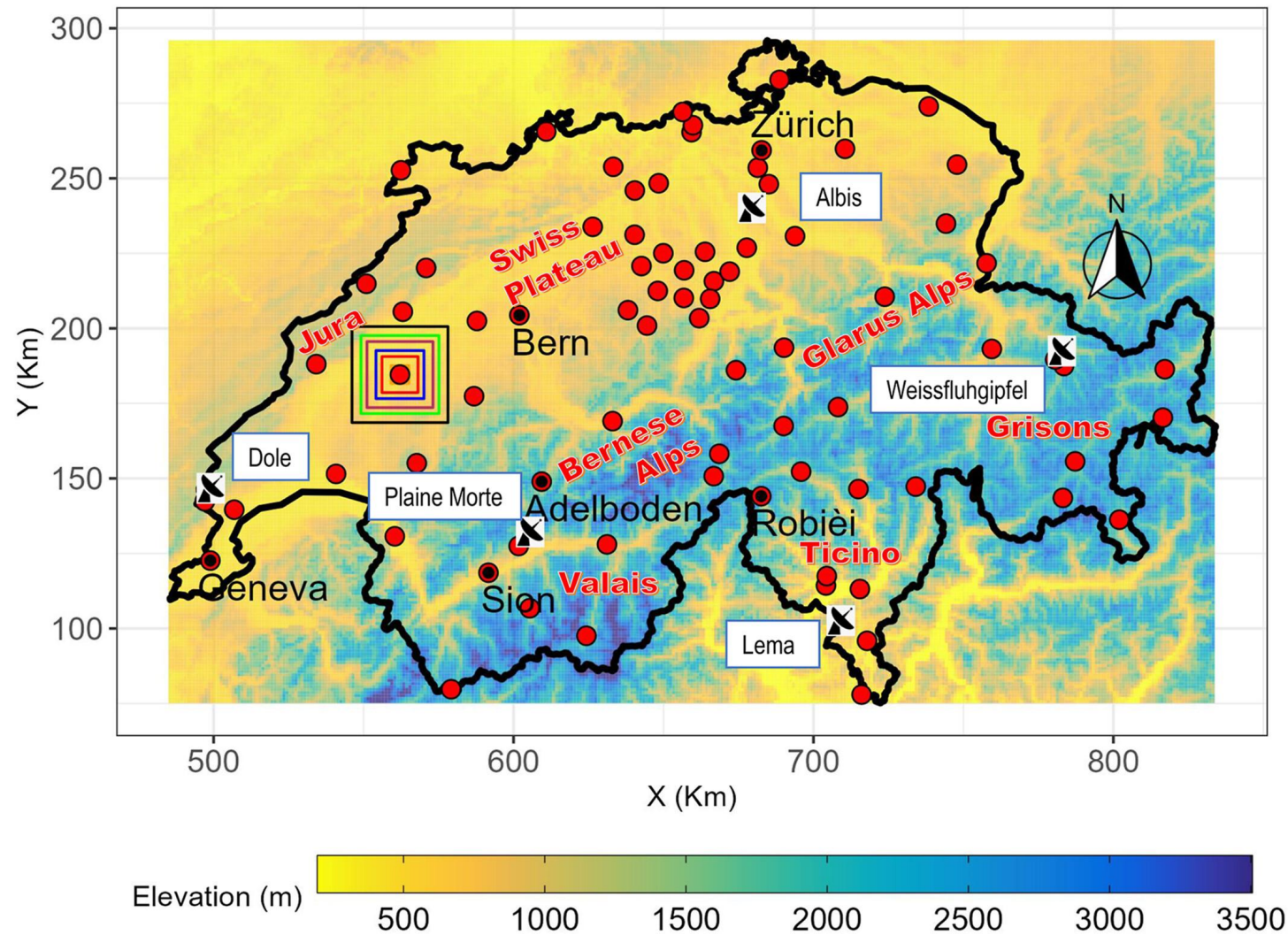
Color Palette



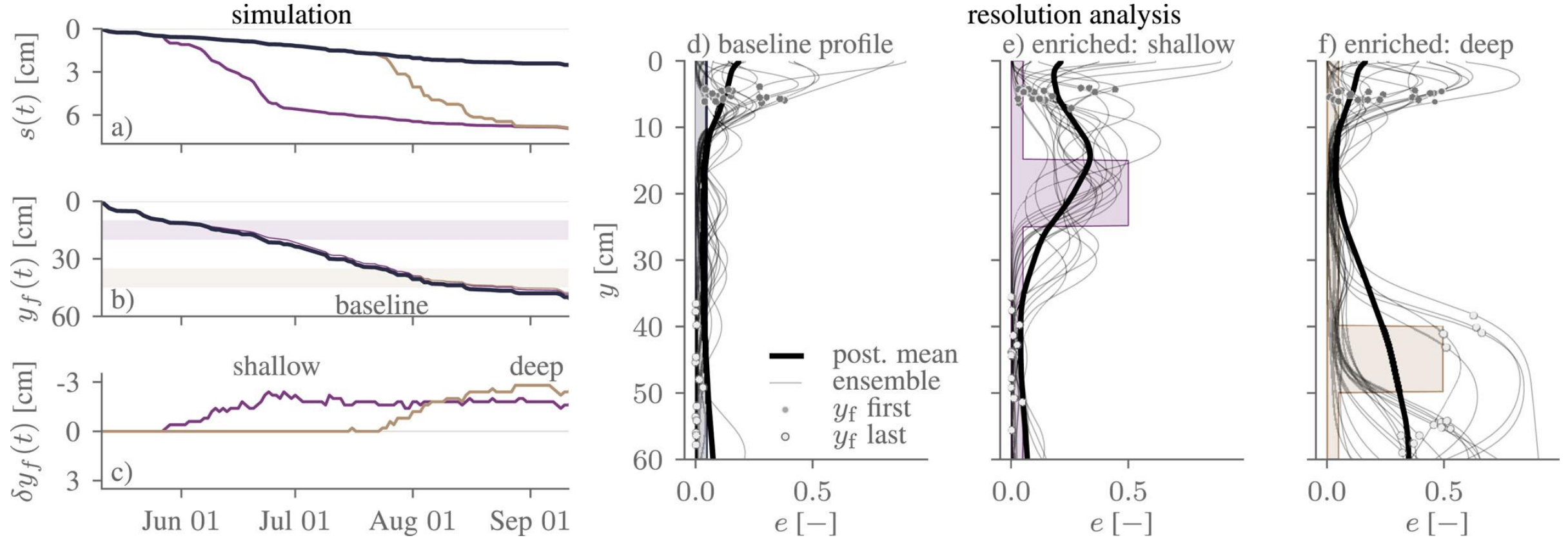
A random sample of figures from recent WRR papers

What are things you like about these figures?

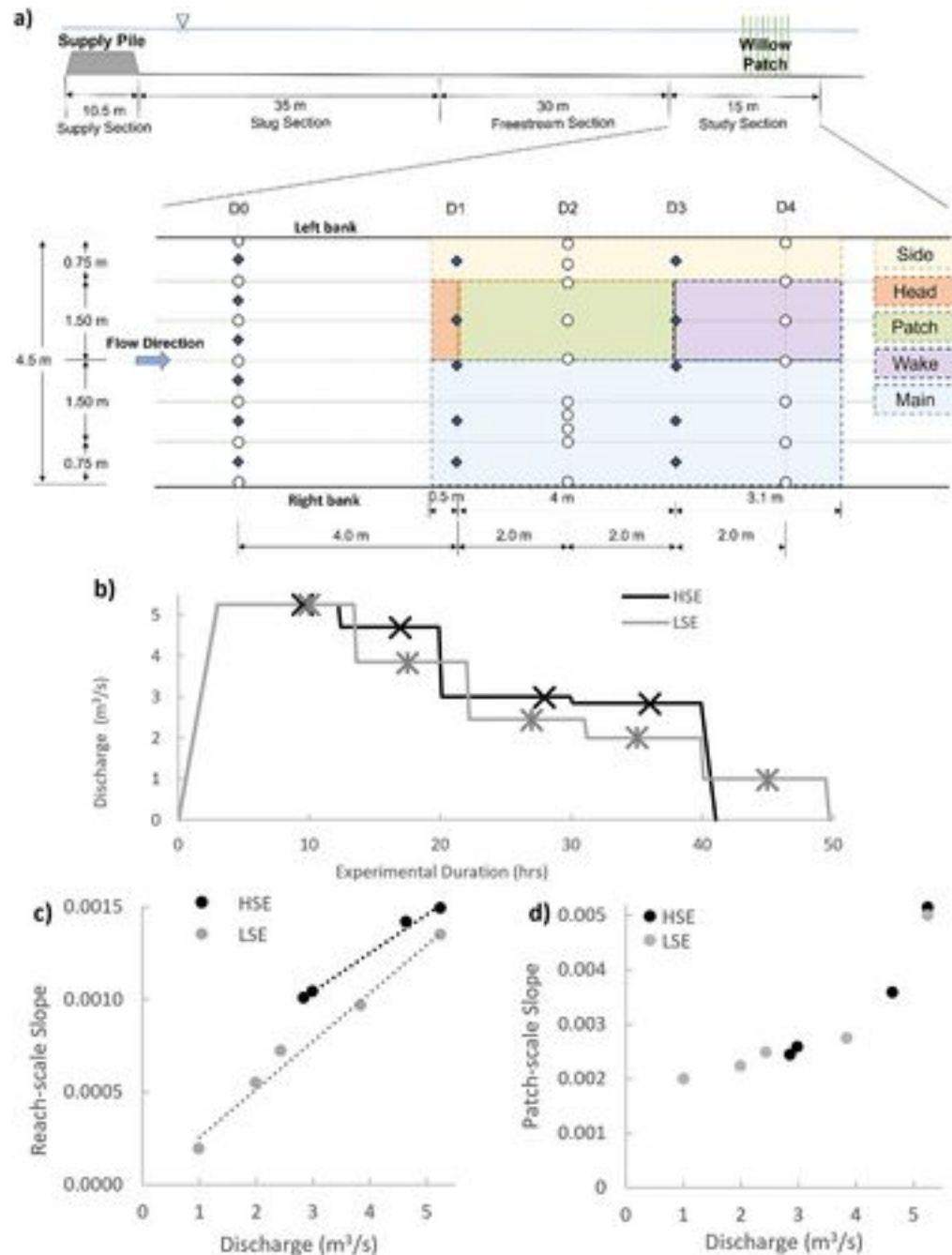
What are changes that could make them better?



Map of Switzerland, the study area. The background color denotes the elevation (meters) above mean sea level. The Radar symbols show the location of the five radars in Switzerland, with their names in the white boxes. The name of some cities is shown in black and the name of some mountains and regions are shown in red. The colored embedded squares show exemplarily the extent of 7 out of the 10 square windows used for data aggregation. They cover areas of 25, 49, 169, 289, 529, 729, and 1,089 km², from the innermost to the outermost. The red colored circles show the location of 79 rain gauges used for evaluation of the gridded product (CPC).



(a–c) Sensitivity of modeled subsidence to excess ice profiles. Compared to the baseline with uniform $e = 0.05$, the shallow and deep scenarios featured ice-enriched layers indicated by the shaded bands in panel (b). For the shallow layer shown in purple, subsidence in panel (a) increased in late May when the thaw front hit the enriched layer below 15 cm shown in panel (b), followed by a sustained reduction in thaw depth relative to baseline, plotted in panel (c). (d–f) Resolution analysis with synthetic observations. The prescribed excess ice profile is shaded, the posterior mean shown with a thick line, and the highest-weight ensemble members with thin lines, with markers to indicate y_f at the time of the first and last observation.



(a) Side and plan views of experimental setup of channel sections and zones. In the plan view, the color-coded areas delineate the different zones within the study section and the solid black lines identify the boundary between the flume bed and side walls. Diamonds indicate bedload sampling locations, and circles show Acoustic Doppler Velocimeter (ADV) measurements. (b) Experimental hydrographs for the high-supply experiment and low-supply experiment with the X's identifying when ADV and sediment sampling began. Water surface slopes over each hydrograph for (c) the entire flume test section and (d) the study section near the patch.